

* Beyond nearby:

1) Line width:

• Tully-Fisher

• Faber-Jackson

→ Measure total velocity width of a galaxy

$$\left[\begin{array}{l} W \sim 2V_{\max} \sin(i) \\ V_{\text{obs}} = V_{\text{true}} \sin(i) \\ \frac{b}{a} = \cos(i) \end{array} \right.$$

Hubble:

• Distant galaxies show redshifts

$$\underbrace{V}_{\substack{\text{(Redshift)} \\ \text{vel.}}} = H_0 \underbrace{D}_{\substack{\text{p.d.} \\ \text{(Hubble's)} \\ \text{const}}} \quad \nwarrow \substack{\text{(Distance to)} \\ \text{object}}$$

$$V_{\text{rad}} = H_0 D + V_{\text{pec}}$$

$$V_{\text{pec}} \approx 10^3 - 10^5 \text{ km/s}$$

→ Due to flows towards local matter: grav

H_0 : Present Day H.C.

$$[H_0] = \frac{1}{s}$$

* $1/H_0 \approx$ Age of the universe (assuming $dH/dt = 0$)

* Works if $v_{pec} \ll H_0 D$:
→ Define the redshift

$$z = \frac{\Delta \lambda}{\lambda}$$

• Nearby: ($z \ll 1$)
 $v = H_0 D = cz \Rightarrow$

$$v = cz$$
$$D = \frac{cz}{H_0}$$

* Current measurements: $H_0 \approx 73.2 \pm 1.3 \text{ km s}^{-1} \text{ Mpc}^{-1}$

